

Appendix: Full Software Dependency Manifest

This appendix lists every library and sub-dependency present in the virtual environment during the project development. The environment was captured using `pip freeze` to ensure maximum reproducibility of the flight delay prediction models.

Table 1: Complete Dependency List (Python 3.9+ Environment)

Package	Version	Package	Version	Package	Version
anyio	4.13.0	fqdn	1.5.1	numpy	2.4.3
appnope	0.1.4	h11	0.16.0	overrides	7.7.0
argon2-cffi	25.1.0	httpcore	1.0.9	packaging	26.0
argon2-bindings	25.1.0	httpx	0.28.1	pandas	3.0.1
arrow	1.4.0	idna	3.11	pandocfilters	1.5.1
asttokens	3.0.1	ipykernel	7.2.0	parso	0.8.6
async-lru	2.3.0	ipython	9.10.0	pexpect	4.9.0
attrs	26.1.0	ipy-pygments	1.1.1	pickleshare	0.7.5
babel	2.18.0	isoduration	20.11.0	pillow	12.1.1
backcall	0.2.0	jedi	0.19.2	platformdirs	4.9.4
beautifulsoup4	4.14.3	Jinja2	3.1.6	plotly	5.24.1
bleach	6.3.0	joblib	1.5.3	prometheus-cl.	0.24.1
certifi	2026.2.25	json5	0.13.0	prompt-toolkit	3.0.52
cffi	2.0.0	jsonpointer	3.1.1	psutil	7.2.2
charset-norm.	3.4.6	jsonschema	4.26.0	ptyprocess	0.7.0
comm	0.2.3	jsonschema-sp.	2025.9.1	pure_eval	0.2.3
contourpy	1.3.3	jupyter-events	0.12.0	pycparser	3.0
coverage	7.13.5	jupyter-lsp	2.3.0	Pygments	2.19.2
cycler	0.12.1	jupyter-client	8.8.0	pyparsing	3.3.2
debugpy	1.8.20	jupyter-core	5.9.1	py-dateutil	2.9.0
decorator	5.2.1	jupyter-server	2.17.0	py-json-logger	4.0.0
defusedxml	0.7.1	j-server-term.	0.5.4	PyYAML	6.0.3
executing	2.2.1	jupyterlab	4.5.6	pyzmq	27.1.0
fastjsonschema	2.21.2	jlab-pygments	0.3.0	referencing	0.37.0
fonttools	4.62.1	jlab-server	2.28.0	requests	2.33.0
kiwisolver	1.5.0	rfc3339-val.	0.1.4	rfc3986-val.	0.1.1
lark	1.3.1	rfc3987-syntax	1.1.0	rpds-py	0.30.0
MarkupSafe	3.0.3	scikit-learn	1.8.0	scipy	1.17.1
matplotlib	3.10.8	seaborn	0.13.2	Send2Trash	2.1.0
matplotlib-inl.	0.2.1	streamlit	1.55.0	six	1.17.0
mistune	3.2.0	sniffio	1.3.1	soupsieve	2.8.3
nbclient	0.10.4	stack-data	0.6.3	terminado	0.18.1
nbconvert	7.17.0	threadpoolctl	3.6.0	tinycss2	1.4.0
nbformat	5.10.4	tornado	6.5.5	traitlets	5.14.3
nest-asyncio	1.6.0	py-dateutil-t.	2.9.0	typing-ext.	4.15.0
notebook	7.5.5	tzdata	2025.3	uri-template	1.3.0
notebook-shim	0.2.4	urllib3	2.6.3	wcwidth	0.6.0
webcolors	25.10.0	webencodings	0.5.1	websocket-cl.	1.9.0